

Highlights

Goal-Oriented Constitutive Transfer Operators for Parametric FFT Homogenization

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- Homogenized stiffness is treated as a static constitutive transfer map.
- Six-load tangential Ritz enrichment targets $D - B^T K^{-1} B$, not all voxel fields.
- A Ritz/Schur identity gives Loewner monotonicity and a residual error bound.
- At fixed rank, two Fourier kernels remove voxel-count dependence online.
- The reduced operator supports screening and rank-revealing conditional inversion.

Goal-Oriented Constitutive Transfer Operators for Parametric FFT Homogenization

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ABSTRACT

Repeated computational homogenization is formulated as a many-query constitutive transfer problem for a fixed microstructure $\mathcal{G}$ and varying constituent vector ξ . Its discrete full-order response is the Schur map $C_{\text{FOM}}^{\text{eff}} = D - B^T K^{-1} B$. We compile the geometry-dependent affine blocks once and enrich a Ritz space only with the six physical macroscopic-strain responses at selected materials. The resulting operator preserves the variational hierarchy and interpolates the effective tensor and its affine sensitivities at enrichment points. A transverse-longitudinal factorization of an isotropic Fourier reference compiles all six-output residual pairings into BB , BK , and KK contractions. Consequently, the matrix error bound is evaluated online by reduced dense algebra, with no voxel-count dependence at fixed rank. For three-dimensional periodic short-fiber composites, a rank-168 operator reaches a maximum relative tensor error of 4.15×10^{-5} over 128 independent materials. On 16 high-precision points, the direct bound has median effectivity 8.94; adding the exact $48 \rightarrow 168$ Schur drop to the bounded enriched tail gives zero upper-bound violations and median effectivity 1.10. The compiled operator then performs material screening, sensitivity-guided conditional identification, and 200 000 uncertainty-propagation queries without further full-order solves.

1. Introduction

Short-fiber composites offer a large design space because constituent properties, fiber content, orientation, aspect ratio, and spatial organization can all be adjusted. Their macroscopic behavior, however, cannot generally be read from the constituent data alone. It must be obtained from a periodic microscopic boundary-value problem whose solution resolves the interaction between material contrast and the heterogeneous geometry. Computational homogenization provides this connection and returns the effective stiffness needed by structural analysis and material design.

The difficulty changes character when homogenization is embedded in a design workflow. A single accurate cell solve is no longer the final task. Material screening, inverse identification, optimization, and uncertainty propagation may require thousands or hundreds of thousands of evaluations for the same microstructure. In three-dimensional elasticity, every candidate normally requires six independent macroscopic load cases on a voxel grid with hundreds of thousands or millions of unknowns. Repeating that calculation while only the constituent coefficients change makes the microscopic solver the dominant computational bottleneck.

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FFT homogenization reduces the cost of each periodic solve, while POD, reduced-basis homogenization, and parametric model reduction compress families of microscopic states [6, 10, 16, 17, 21, 28]. Nevertheless, the engineering quantity ultimately queried is usually not the entire fluctuation field. It is the 6×6 effective constitutive tensor, and in identification problems also its derivatives with respect to the material parameters. A state-oriented approximation can therefore spend offline information on field directions that have little influence on this small output. Moreover, a useful many-query model needs an error indicator that can be evaluated without invoking the voxel fields or an FFT solve for every candidate.

This paper asks whether the fixed microstructure can instead be compiled into a reusable constitutive transfer operator. The proposed answer starts from the Schur-complement structure of linear computational homogenization. The geometry-dependent operators are assembled once; subsequent enrichment uses only the six physical responses associated with the macroscopic strain space. The reduced model is then constructed to preserve the Ritz energy hierarchy, to interpolate constitutive sensitivities at selected materials, and to expose a residual identity directly associated with the effective stiffness.

The central methodological step is an offline–online error estimator. An isotropic reference inverse separates into transverse and longitudinal Fourier kernels, allowing all voxel-level residual contractions to be precomputed. Online evaluation consequently involves only reduced dense matrices. This connects the physical six-load enrichment, the constitutive output, and the error-control mechanism in one transfer formulation rather than treating them as independent accelerations.

The specific methodological contributions are:

- (i) a geometry-compiled, six-load Ritz transfer operator for the effective Schur map, with constitutive and sensitivity interpolation at enrichment points;
- (ii) a matrix-valued residual compiler based on precomputed BB , BK , and KK contractions and two isotropic Fourier kernels, making online error control independent of the voxel count at fixed rank;
- (iii) causal tensor- and sensitivity-oriented enrichment rules together with a truth-free relative certificate stopping criterion; and
- (iv) a reproducible assessment spanning high-precision energetic checks, equal-budget controls, voxel and morphology variation, and three many-query applications.

Several properties used to analyze this construction are deliberately not claimed as new. Galerkin best approximation, compliant-output quadratic error, coercivity-scaled residual bounds, and affine offline–online residual norms are standard RB results [3, 22, 23]. POD/RB acceleration of constitutive homogenization across linear and nonlinear

settings is likewise well established [6, 10, 17, 19, 21, 28]. Reduced Schur systems, Schur-complement perturbation bounds, and their certification in static-condensation RB methods are also established [5, 12, 24]. Likewise, projection-based interpolation of parameter derivatives is established in parametric model reduction [2], and variational bounds for homogenized operators predate this work [15, 27]. We state these classical results only in their matrix-valued constitutive specialization and cite their proofs. The novelty claim is therefore deliberately narrow. To the authors' knowledge, the closest antecedents do not compile all six-output residual pairings of the homogenized constitutive map through the transverse/longitudinal Fourier split into the dense BB , BK , and KK arrays to obtain a voxel-free online matrix bound integrated with physical tangential enrichment and constitutive sensitivity interpolation.

Table 1 positions this combination against the closest antecedents. POD, reduced bases, CRE certification, affine separation, and FFT reduction are not claimed as isolated novelties.

The remainder of the paper follows the mathematical construction. Section 2 introduces periodic homogenization and its Schur-complement transfer representation. Section 3 constructs the goal-oriented reduced operator, Section 4 derives its offline–online Schur estimator, and Section 5 establishes the constitutive sensitivity interpolation property. Section 6 defines the FFT implementation and numerical protocol. Results and their interpretation are presented together in Section 7, followed by the conclusions in Section 8.

2. Periodic homogenization and constitutive transfer

2.1. Periodic cell problem and effective response

Let Y be a periodic representative cell with local stiffness $\mathbb{C}(\mathbf{x}; \xi, \mathcal{G})$. For a prescribed macroscopic strain $\bar{\epsilon}(g)$, represented by the Mandel vector $g \in \mathbb{R}^6$, the fluctuation u_g belongs to the periodic, zero-mean compatible space and satisfies

$$\int_Y \epsilon(v) : \mathbb{C}(\mathbf{x}; \xi, \mathcal{G}) : [\bar{\epsilon}(g) + \epsilon(u_g)] \, d\mathbf{x} = 0 \quad \forall v \in \mathcal{V}_{\text{per}}, \quad (1)$$

where

$$\epsilon(u) = \frac{1}{2} (\nabla u + \nabla u^T), \quad \langle u_g \rangle_Y = 0.$$

The effective constitutive tensor is defined by the volume-averaged stress,

$$C^{\text{eff}}(\xi; \mathcal{G})g = \langle \mathbb{C}(\mathbf{x}; \xi, \mathcal{G}) : [\bar{\epsilon}(g) + \epsilon(u_g)] \rangle_Y. \quad (2)$$

Table 1

Proposition-level positioning against direct methodological antecedents. The middle column identifies what is inherited; the last column states the remaining boundary of the present contribution.

Reference	Established antecedent used here	Boundary relative to the present work
Prud'homme et al. [22] and Quarteroni et al. [23]	Galerkin best approximation, compliant-output error, coercivity-scaled residual bounds, and affine offline–online evaluation	Not homogenization-specific and not a six-output Fourier residual compiler
Boyaval [3]	Reduced-basis homogenization with output error control and offline–online separation	Scalar elliptic homogenized outputs rather than the present 3D constitutive transfer realization
Yvonnet and He [28], Fritzen and Leuschner [6], Hernández et al. [10], Michel and Suquet [19], and Kunc and Fritzen [17]	POD/RB, variational reduction, and hyperreduced computational homogenization for constitutive response	Establish that reduced, structure-preserving constitutive homogenization is not new; they do not provide the present linear-parametric six-output Fourier bound compiler
Baur et al. [2]	Parameter value, gradient, and Hessian interpolation under projection-space conditions	Supplies the classical interpolation principle, not the static six-load constitutive specialization
Kamiński [14], Guo et al. [7], and Lukeš and Rohan [18]	Sensitivity of homogenized properties, reduced constitutive surrogates with derivatives, and sensitivity-guided hyperelastic reduction	Establish that sensitivities and inverse/design use are not new; they do not provide exact six-load snapshot-containment interpolation with the present bound compiler
Huynh et al. [12], Eftang and Patera [5], and Smetana [24]	Reduced Schur systems, Schur-complement perturbation error, port reduction, and static-condensation certification	Different component/interface problem; no constitutive FFT two-kernel BB/BK/KK compilation
Huynh et al. [13] and Hain et al. [8]	SCM coercivity lower bounds and hierarchical estimators based on nested reduced approximations	Classical alternatives/ingredients; the present realization adds an exact matrix Schur drop to a compiled six-output tail bound
Kerfriden et al. [15] and Hoang et al. [11]	CRE and goal-oriented/two-field certification for homogenization or affinely parametrized elasticity	Different equilibrated two-field estimator family
Moulinec and Suquet [20] and Vondrejč et al. [26, 27]	Isotropic Fourier Green operators, Fourier–Galerkin homogenization, and discretization bounds	No reduced six-output residual compiler or ROM-error bound of the form proposed here
Kochmann et al. [16], Vondrejč et al. [25], and Hauck et al. [9]	FFT acceleration/reduction by sparse frequency sampling, low-rank tensors, and tensor-train Fourier transforms	Reduce the Fourier solve or field representation rather than compiling a parametric constitutive Schur map and its online matrix bound
Pham and Lee [21]	Reduced-basis thermal and elastic periodic homogenization with offline–online corrector gradients	No matrix-valued residual certification or two-kernel dense online estimator
Present method	Uses the classical results above	Compiles all affine residual pairings for six constitutive outputs into transverse/longitudinal BB/BK/KK contractions and couples them to physical causal enrichment

Equations (1)–(2) make the separation central to the paper explicit: $\mathcal{G}$ fixes the phase and orientation fields, while ξ changes the constituent law over repeated queries.

2.2. Discrete Schur-complement transfer representation

Let $u \in \mathbb{R}^n$ denote the compatible, zero-mean fluctuation field and $g \in \mathbb{R}^6$ a macroscopic Mandel strain vector. After discretizing a linear periodic cell problem, the microscopic energy is

$$\mathcal{E}(u, g; \xi) = \frac{1}{2} u^T K(\xi) u - u^T B(\xi) g + \frac{1}{2} g^T D(\xi) g. \quad (3)$$

Stationarity of Eq. (3) with respect to the fluctuation, applied to the six independent macroscopic strains, gives the full-order response matrix $X(\xi) \in \mathbb{R}^{n \times 6}$:

$$K(\xi) X(\xi) = B(\xi), \quad (4)$$

Eliminating this response from Eq. (3) then produces the effective stiffness as the Schur complement

$$\boxed{C_{\text{FOM}}^{\text{eff}}(\xi) = D(\xi) - B(\xi)^T K(\xi)^{-1} B(\xi)}. \quad (5)$$

Hereafter, $C_{\text{FOM}}^{\text{eff}}$ denotes this converged discrete FFT response, whereas C^{eff} is reserved for the continuum quantity defined in Eq. (2). The ROM bound controls $C_r^{\text{eff}} - C_{\text{FOM}}^{\text{eff}}$; voxel refinement is reported separately. The formulation is first abstract: $K(\xi)$ is symmetric positive definite on the compatible fluctuation space and B contains the six macroscopic right-hand sides. Section 6 defines the short-fiber FFT realization of this structure.

2.3. Material parameters and geometry-affine separation

The physical material vector used throughout the study is

$$\xi = (E_m, \nu_m, E_L, E_T, G_{LT}, \nu_{LT}, \nu_{TT}) \in \mathcal{D}_\xi.$$

It contains the isotropic matrix Young modulus and Poisson ratio together with five engineering constants for the transversely isotropic fiber. We retain this dimensional seven-parameter interface because it is the one used to generate and validate all numerical evidence in the paper. Linear elasticity admits a reduction by the global elastic scale E_m , as discussed in Appendix E, but that optional quotient is not used in the reported campaigns.

For the isotropic-matrix/transversely-isotropic-fiber case, the local elastic law is affine in seven coefficients

$$\gamma(\xi) = (\lambda_m, \mu_m, c_{TT}, c_{TT}^X, c_{LT}, c_{LL}, g_{LT}),$$

Table 2

Admissible constituent-property box for the fixed-geometry transfer operator.

Parameter	Minimum	Maximum
Em	1.1000	4.4000
nu_m	0.3500	0.4200
Ef_L	72.0000	395.0000
Ef_T	6.0000	23.0000
G_LT	8.0000	30.0000
nu_LT	0.2000	0.2600
nu_TT	0.3500	0.4000

so that

$$\begin{aligned}
 K(\xi) &= \sum_{q=1}^Q \gamma_q(\xi) K_q(\mathcal{G}), & B(\xi) &= \sum_{q=1}^Q \gamma_q(\xi) B_q(\mathcal{G}), \\
 D(\xi) &= \sum_{q=1}^Q \gamma_q(\xi) D_q(\mathcal{G}), & Q &= 7.
 \end{aligned} \tag{6}$$

The microstructure $\mathcal{G}$ contains phase, orientation, periodicity, discretization, fiber volume fraction, aspect ratio, and topology. The operators $K_q(\mathcal{G})$, $B_q(\mathcal{G})$, and $D_q(\mathcal{G})$ are geometry-dependent and are assembled once. The online material dependence is carried only by $\gamma_q(\xi)$. The separation in Eq. (6) is the mechanism that turns the Schur map in Eq. (5) into an offline–online transfer problem.

Table 2 specifies the physical parameter box used for all training, validation, and many-query campaigns.

3. Goal-oriented constitutive transfer reduction

The construction has one offline stage and one online stage. Offline, the continuous microstructure is rasterized, its affine geometry operators are assembled, selected material points are solved under the six physical loads, and the reduced and residual-contraction tensors are compiled. Online, a new physical material vector is converted into seven affine coefficients and only small dense systems are evaluated:

$$\underbrace{\mathcal{G} \longrightarrow \{K_q, B_q, D_q\}_{q=1}^7 \longrightarrow V_r \longrightarrow \{BB, BK, KK\}}_{\text{offline compilation}} \quad \underbrace{\xi \longrightarrow \gamma \longrightarrow (C_r^{\text{eff}}, \nabla_{\xi} C_r^{\text{eff}}, \eta_{\text{cert}})}_{\text{online query}}. \tag{7}$$

Equation (7) also fixes the information boundary: voxel fields are used during compilation, whereas a deployed query uses only the saved reduced arrays. Figure 1 makes this separation concrete using the dimensions of the final operator: 28 six-load material responses produce a rank-168 space, while the online path receives only a seven-parameter vector and the compiled reduced arrays.

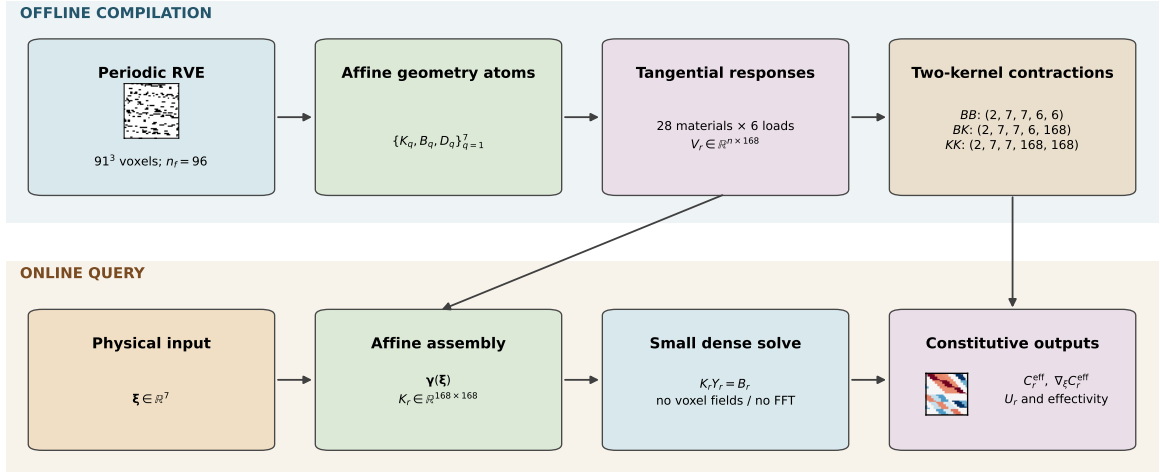


Figure 1: Data flow of the implemented constitutive-transfer method. The upper band contains the voxel-dependent offline operations and the actual dimensions of the two-kernel contractions. The lower band contains the online query, which assembles and solves only a 168×168 dense system and returns the effective tensor, its material sensitivities, and the Schur bound.

The reported basis is stopped at the scientific target 10^{-4} . During method validation this decision is checked on an untouched material set; once the estimator is compiled, Eq. (24) provides the corresponding truth-free online stopping test. Smaller errors are retained for solver and identity diagnostics but do not motivate further enrichment.

3.1. Ritz projection and six-load tangential enrichment

Let $V_r \in \mathbb{R}^{n \times r}$ be a reduced Ritz basis built from selected full-order response matrices. Projection gives

$$K_r(\xi) = V_r^T K(\xi) V_r, \quad B_r(\xi) = V_r^T B(\xi),$$

$$X_r(\xi) = V_r K_r(\xi)^{-1} B_r(\xi),$$

and substitution into the full Schur map (5) gives the reduced transfer output

$$\boxed{C_r^{\text{eff}}(\xi) = D(\xi) - B_r(\xi)^T K_r(\xi)^{-1} B_r(\xi).} \quad (8)$$

Each enrichment material $\xi^{(j)}$ contributes the six physical response columns from Eq. (4),

$$X(\xi^{(j)}) = K(\xi^{(j)})^{-1} B(\xi^{(j)}),$$

and the basis is updated as

$$V_{r+6} = \text{orth} [V_r, X(\xi^{(j)})]. \quad (9)$$

Thus, one material point corresponds transparently to six full-order load solves and at most six new reduced directions. We report the number of material points, load solves, and reduced dimension separately in Section 7.

The adaptive material-space loop is output-oriented:

$$\xi^* = \arg \max_{\xi \in \Xi_{\text{cand}}} \eta_C(\xi), \quad (10)$$

where η_C is the tensor-oriented score assembled from the Schur estimator derived in Section 4. Only the maximizer ξ^* in Eq. (10) is sent to the full-order FFT solver for the next six-load tangential block through Eq. (9).

4. Offline–online Schur error estimation

4.1. Residual energy and two-kernel factorization

Define the residual matrix

$$R(\xi) = B(\xi) - K(\xi)X_r(\xi). \quad (11)$$

A field residual alone is not the target quantity of interest. We instead map Eq. (11) back to the six-output constitutive space through the positive-semidefinite matrix

$$\mathcal{E}_M(\xi) = R(\xi)^T M(\xi)^{-1} R(\xi),$$

where M is an isotropic Fourier reference operator. For every nonzero wave vector, its inverse separates exactly into transverse and longitudinal traction kernels. This Green-operator decomposition is classical [20, 26]; the proposition below concerns the subsequent pairwise compilation, not the decomposition itself:

$$M^{-1} = \mu_0^{-1} P_T + (\lambda_0 + 2\mu_0)^{-1} P_L. \quad (12)$$

Pairwise Gram precomputation for affine residual norms is also standard in offline–online RB methods [22, 23]. The construction below specializes that principle simultaneously to the six constitutive columns and to the two physical Fourier kernels. Using the decomposition (12), and writing $X_r = V_r Y_r$ and $R = \sum_p \gamma_p (B_p - K_p V_r Y_r)$, we precompute,

for $\alpha \in \{T, L\}$,

$$BB_{pq}^\alpha = B_p^T P_\alpha B_q, \quad BK_{pq}^\alpha = B_p^T P_\alpha K_q V_r, \quad KK_{pq}^\alpha = V_r^T K_p^T P_\alpha K_q V_r. \quad (13)$$

The online kernel matrix is then assembled as

$$E_\alpha = \sum_{p,q} \gamma_p \gamma_q (BB_{pq}^\alpha - BK_{pq}^\alpha Y_r - Y_r^T (BK_{qp}^\alpha)^T + Y_r^T K K_{pq}^\alpha Y_r), \quad (14)$$

$$R^T M^{-1} R = \mu_0^{-1} E_T + (\lambda_0 + 2\mu_0)^{-1} E_L. \quad (15)$$

Proposition 1 (Exact six-output two-kernel compilation). *Under the affine residual decomposition above and an isotropic reference operator, Eqs. (13)–(15) recover $R^T M^{-1} R$ exactly, up to floating-point arithmetic. After the reduced solve, their evaluation costs $O(Q^2 r^2 + Q^2 r n_g)$ operations with $n_g = 6$, stores $O(Q^2 r^2)$ numbers, and has no explicit voxel-count dependence for fixed r .*

Including affine reduced assembly and a dense factorization, one transfer query costs $O(Qr^2 + r^3 + Q^2 r^2)$ for fixed $n_g = 6$. A hierarchical query repeats this work for its two ranks; neither expression depends on N_{vox} once those ranks have been compiled.

Appendix C proves Proposition 1. The compiled contractions are stored as `online_estimator.npz` beside the reduced operators; direct-Fourier and dense evaluations are compared automatically during compilation.

4.2. Ritz/Schur identity and Loewner hierarchy

The following statement is the standard Galerkin energy identity for a compliant output, written columnwise for the matrix-valued constitutive Schur map [22, 23]. Its use in homogenization and the associated energetic bounds are also classical [3, 15]; reduced Schur complements and their error control also appear in static-condensation RB methods [5, 12, 24]. We record the specialization because it exposes the six-output residual matrix used by the proposed compiler; the identity itself is not a novelty claim.

For $K(\xi)$ symmetric positive definite on the compatible fluctuation space and X_r the Ritz projection of $X = K^{-1} B$ onto V_r , the cited classical identity gives

$$\Delta C_r(\xi) := C_r^{\text{eff}}(\xi) - C_{\text{FOM}}^{\text{eff}}(\xi) = (X - X_r)^T K (X - X_r) = R^T K^{-1} R \geq 0. \quad (16)$$

For nested spaces $V_r \subset V_s$, the same classical variational restriction gives

$$C_r^{\text{eff}}(\xi) \geq C_s^{\text{eff}}(\xi) \geq C_{\text{FOM}}^{\text{eff}}(\xi), \quad (17)$$

a Loewner-monotone constitutive hierarchy. This is stronger than scalar error decay: the reduced stiffnesses approach the full-order stiffness from the energy-upper side.

4.3. Beta-scaled bound and online reference selection

If the reference operator satisfies

$$K(\xi) \geq \beta(\xi)M(\xi), \quad \beta(\xi) > 0, \quad (18)$$

then

$$0 \leq \Delta C_r(\xi) \leq U_r(\xi) := \beta(\xi)^{-1} R(\xi)^T M(\xi)^{-1} R(\xi), \quad (19)$$

and

$$\|C_r^{\text{eff}}(\xi) - C_{\text{FOM}}^{\text{eff}}(\xi)\|_F \leq \eta_{\text{cert}}(\xi) := \|U_r(\xi)\|_F. \quad (20)$$

The matrix inequality (19) is the operator form of the standard coercivity-scaled RB residual bound [22, 23]; Eq. (20) is its scalar reporting form. The contribution here is not the coercivity argument, but its simultaneous evaluation for all six constitutive outputs by the compiled two-kernel contractions. For validation only, its scalar effectivity index is

$$\mathcal{I}_{\text{eff}} = \frac{\eta_{\text{cert}}}{\|C_r^{\text{eff}} - C_{\text{FOM}}^{\text{eff}}\|_F}, \quad (21)$$

which is at least one for a valid upper bound. It measures sharpness but is not used by the online stopping rule.

The direct bound can be sharpened without changing its reference operator. For two nested Ritz spaces $V_r \subset V_s$, define the exactly computable reduced Schur drop $D_{r,s} = C_r^{\text{eff}} - C_s^{\text{eff}}$. If U_s is any valid matrix upper bound for the enriched tail $C_s^{\text{eff}} - C_{\text{FOM}}^{\text{eff}}$, then the hierarchy in Eq. (17) gives

$$0 \leq D_{r,s} \leq \Delta C_r = D_{r,s} + \Delta C_s \leq D_{r,s} + U_s =: U_{r \leftarrow s}^{\text{hier}}. \quad (22)$$

Hierarchical reduced estimators based on two nested approximations are established in model reduction [8]. Here no saturation assumption is needed because the remaining enriched error is not discarded: it is bounded by the same coercivity-scaled two-kernel construction. Evaluation requires two small reduced solves and the compiled estimator at level s , but still contains no voxel field or online FFT.

For a stopping criterion that remains meaningful when the observed error is very small, define the computable relative certificate width

$$\delta_r(\xi) = \frac{\|U_r(\xi)\|_F}{\|C_r^{\text{eff}}(\xi)\|_F}. \quad (23)$$

Here U_r may be the direct matrix bound in Eq. (19) or the hierarchical bound in Eq. (22). Because $\|C_{\text{FOM}}^{\text{eff}}\|_F \geq (1 - \delta_r)\|C_r^{\text{eff}}\|_F$, every $\delta_r < 1$ gives

$$\frac{\|C_r^{\text{eff}} - C_{\text{FOM}}^{\text{eff}}\|_F}{\|C_{\text{FOM}}^{\text{eff}}\|_F} \leq \frac{\delta_r}{1 - \delta_r}. \quad (24)$$

Thus the discrete-ROM target 10^{-4} is guaranteed whenever $\delta_r \leq 10^{-4}/(1+10^{-4})$. This rule uses no full-order validation error and is therefore also the causal stopping rule for auxiliary enrichment. Effectivity remains a sharpness diagnostic, but it is not used as a stopping quantity when its true-error denominator lies far below the adopted accuracy floor.

For each material, 129 reproducible isotropic reference shapes are tested and the valid reference minimizing the Frobenius bound is selected. The coercivity factor is the minimum generalized eigenvalue over the matrix and fiber local stiffnesses, with a conservative numerical margin; its derivation is given in Appendix B. The high-precision study in Section 7.2 compares the direct $r = 48$ bound with the hierarchical $48 \rightarrow 168$ bound. SCM coercivity surrogates [13] and anisotropic reference operators are viable ways to tighten the residual tail, but the ablation shows that they are unnecessary to meet the target effectivity here. The bound controls reduction error with respect to the discrete FFT problem, not voxel discretization or the continuum limit.

5. Constitutive sensitivities and tangential Hermite interpolation

Differentiating the reduced Schur map (8) gives the affine coefficient sensitivities

$$\boxed{\frac{\partial C_r^{\text{eff}}}{\partial \gamma_q} = D_q - B_q^T X_r - X_r^T B_q + X_r^T K_q X_r.} \quad (25)$$

Physical derivatives with respect to $(E_m, \nu_m, E_L, E_T, G_{LT}, \nu_{LT}, \nu_{TT})$ follow by the chain rule through $\gamma(\xi)$; the coefficient map is listed explicitly in Appendix A. Sensitivities of homogenized elastic properties and reduced constitutive surrogates with parameter derivatives are established topics [7, 14, 18]; neither sensitivity analysis nor its use in inverse problems is claimed here in isolation.

Parameter-response and parameter-derivative matching under projection-space containment are established results in interpolatory parametric model reduction [2]. Applied to the static symmetric map $D - B^T K^{-1} B$, the cited result

states that if a material point $\xi^{(j)}$ has been used for tangential enrichment and the six full-order response columns $X(\xi^{(j)})$ are contained in V_r , then

$$C_r^{\text{eff}}(\xi^{(j)}) = C_{\text{FOM}}^{\text{eff}}(\xi^{(j)}). \quad (26)$$

Moreover, for every affine coefficient γ_q ,

$$\left. \frac{\partial C_r^{\text{eff}}}{\partial \gamma_q} \right|_{\xi^{(j)}} = \left. \frac{\partial C_{\text{FOM}}^{\text{eff}}}{\partial \gamma_q} \right|_{\xi^{(j)}}. \quad (27)$$

No derivative snapshots are required because the projection-space containment condition is already satisfied by the six physical response columns.

Equations (26)–(27) explain why the six-load snapshots are more than training vectors: they impose a local constitutive Hermite interpolation condition on the Schur-complement transfer map. Sensitivity-aware enrichment can therefore be treated as a separate mode for inverse identification and optimization, where the target quantity is $(C_{\text{FOM}}^{\text{eff}}, \nabla_{\xi} C_{\text{FOM}}^{\text{eff}})$, not only $C_{\text{FOM}}^{\text{eff}}$.

For the causal comparison, let $[a_j, b_j]$ be the interval of parameter ξ_j and let $\eta_C = \|U_r\|_F / \|C_r^{\text{eff}}\|_F$ denote the relative tensor score. The sensitivity score and the mixed selection score are defined by

$$\eta_{\text{VC}}(\xi) = \max_j (b_j - a_j) \left\| \frac{\partial}{\partial \xi_j} \left(\frac{R^T M^{-1} R}{\|C_r^{\text{eff}}\|_F} \right) \right\|_F, \quad \eta_{\text{mix}} = \sqrt{\eta_C^2 + \eta_{\text{VC}}^2}. \quad (28)$$

The optimized reference is used at each evaluation, and the derivative in Eq. (28) is approximated by centered differences with relative step 10^{-3} , using only repeated dense evaluations of the compiled estimator. The factor $b_j - a_j$ removes the units of each physical coordinate. Thus this rule targets rapid variation of the residual-energy matrix without candidate FOM sensitivities, voxel fields, or online FFTs. In the comparison below, standard energy greedy maximizes $\|R^T M^{-1} R\|_F / \|C_r^{\text{eff}}\|_F$, QoI/Schur greedy maximizes η_C , and the sensitivity-aware rule maximizes η_{mix} .

6. FFT implementation and numerical protocol

The full-order baseline is an FFTHomPy/CuPy periodic Fourier–Galerkin solve with six independent 3D macroscopic load cases. Three principal, noninterchangeable profiles are used in the paper: truth computations use `float64` and relative tolerance 10^{-10} , the strict snapshot audit uses `float64` and 10^{-8} , and the economical production basis pass uses the timing profile with `float32` and 10^{-5} . An additional `rom_floor` profile (`float32`, 10^{-4}) is reserved for

Table 3

Fixed short-fiber RVE used throughout the fixed-geometry study.

Quantity	Value
Grid	91 x 91 x 91 voxels
Fiber count	96
Target / actual volume fraction	0.200 / 0.199284
Volume-fraction error	-0.0716 percentage points
Aspect ratio	15.0
Fiber diameter / length	1.0 / 15.0 micrometers
RVE side length	18.060 micrometers
Length/domain ratio	0.830566
Orientation state	quasi-isotropic
Phase/orientation hash prefix	0c167523f0a5 / 23b849c7644a

rapid feasibility pilots and is not used for the final truth comparison. The economical profiles are used only to construct the offline basis; their error is checked against independent truth solves and the strict snapshot profile remains available through `-basis-profile snapshot`. Every solve must declare convergence and records the relative residual of each load, iteration count, symmetry defect, and minimum effective-tensor eigenvalue. Failed, incomplete, non-SPD, or nonconverged solves abort the campaign.

The implementation was checked on a homogeneous cell and an analytical 3D two-phase laminate, including Mandel energy equivalence, compatibility, projected equilibrium, periodicity, and Hill–Mandel consistency. The homogeneous-cell relative error is 4.63×10^{-16} . All 16 independent truth solves used for the bound converged, with maximum relative residual 9.96×10^{-11} and at most 198 CG iterations. A separate refinement uses the same continuous 96-fiber geometry at 48^3 , 64^3 , 91^3 , 128^3 , and a geometry campaign varies the microstructure while keeping the box and 91^3 grid fixed. Table 3 summarizes the baseline discretization, while Figure 2 connects the continuous periodic fibers to their voxelization and realized orientation statistics.

The validation metric is the relative Frobenius tensor error

$$e_{\text{rel}} = \frac{\|C_r^{\text{eff}}(\xi) - C_{\text{FOM}}^{\text{eff}}(\xi)\|_F}{\|C_{\text{FOM}}^{\text{eff}}(\xi)\|_F}. \quad (29)$$

The present study uses 10^{-4} in Eq. (29) as the scientific reporting floor. Values near 10^{-7} or 10^{-8} are retained as algebraic and solver diagnostics, but they do not trigger additional enrichment once the scientific target has been reached. Campaign manifests and manuscript build commands are documented in Appendix F.

The equal-budget comparator is causal by construction. A 4096-point Sobol candidate pool, 256 untouched validation materials, ten random schedules, and budgets of 1, 2, 4, 6, 8, 10, 15, 20 materials are frozen before evaluation. Sobol+POD, random, standard energy greedy, QoI/Schur greedy, and a sensitivity-aware Schur rule each pay six

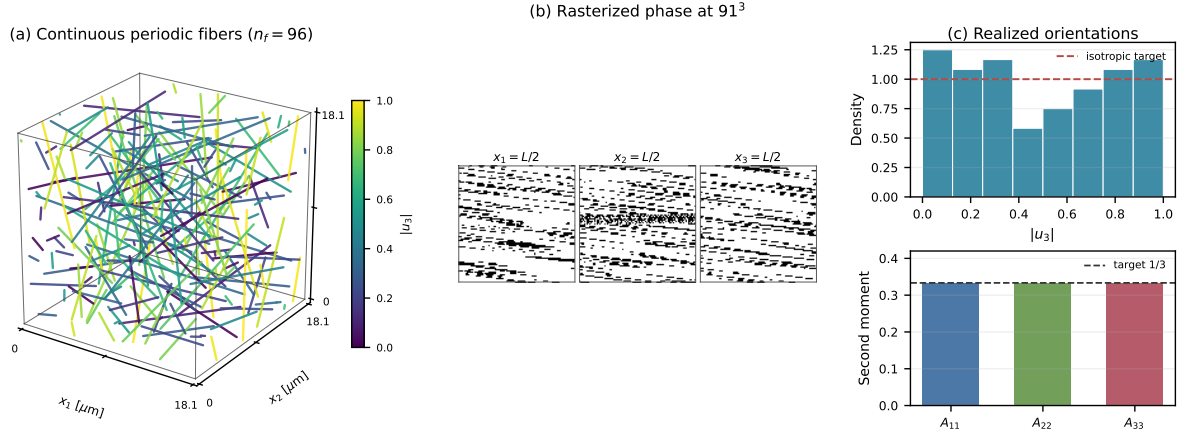


Figure 2: Traceable morphology of the fixed SAM short-fiber RVE. (a) The 96 continuous centerlines are wrapped through the periodic cell and colored by $|u_3|$. (b) Three orthogonal sections show the 91^3 rasterization used by the FFT solver. (c) The realized axial orientation distribution and second-order orientation tensor are compared with the isotropic targets. The same accepted geometry is reused for every material query in the fixed-geometry study.

full-order loads per selected material. Candidate full-order errors are never available to a selection rule. Mean, 95th-percentile, and maximum errors are evaluated on the same 256 truth tensors and supplemented by paired bootstrap intervals. As a nonintrusive control, an RBF predicts the 21 independent entries of the matrix logarithm of C_{FOM}^{eff} in normalized seven-parameter coordinates. Gaussian, inverse-multiquadric, and inverse-quadratic kernels, five shape scales, and four smoothing levels are selected by leave-one-out error using only the available training tensors. Exponentiation guarantees an SPD prediction; this is the classical log-Euclidean construction [1], not a claimed contribution.

7. Results and discussion

The numerical evidence follows the methodological construction. We first establish transfer accuracy and the reliability of the Schur bound, then compare causal enrichment rules and assess constitutive sensitivities. Discretization and geometry checks delimit the claim before the compiled operator is used in three many-query applications. Results and their interpretation are reported together in each subsection.

7.1. ROM convergence

Figure 3 shows the independent 32-point validation curve. The $r = 66$ transfer operator reaches maximum relative tensor error 3.65×10^{-4} . Increasing the nested basis reduces the sampled error, and $r = 168$ reaches 5.20×10^{-6} on the same 32-point set. A larger 128-point validation gives mean 3.87×10^{-6} , 95th percentile 1.48×10^{-5} , and maximum 4.15×10^{-5} . This larger set is reported as the more conservative robustness assessment in Table 4. Because its maximum remains below the 10^{-4} criterion in Eq. (29), it does not alter the stopping decision.

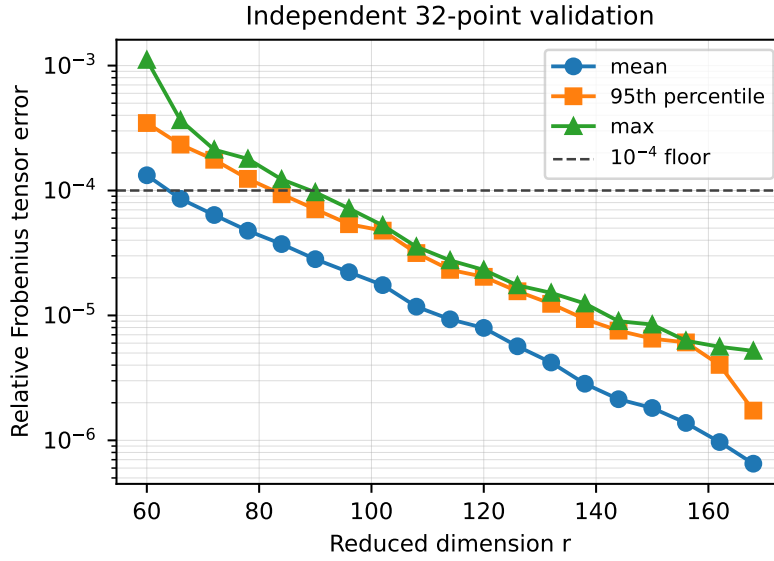


Figure 3: Convergence of the fixed-geometry tangential Ritz operator on the 32-point independent validation set. The dashed line marks the adopted 10^{-4} scientific reporting floor.

Table 4

Independent tensor-validation errors for the fixed geometry.

Case	Samples	Mean	P95	Max	Median online s
$r=66$, 32-point set	32	8.579e-05	2.333e-04	3.650e-04	2.259e-04
$r=168$, same 32-point set	32	6.502e-07	1.730e-06	5.203e-06	9.801e-04
$r=168$, 128-point GPU set	128	3.866e-06	1.477e-05	4.147e-05	1.053e-03

7.2. Offline–online bound and Loewner hierarchy

The dense two-kernel estimator and the direct Fourier implementation agree in `float64`. Their maximum relative discrepancy is 9.12×10^{-12} at $r = 48$; at $r = 168$ it is 2.17×10^{-9} because the reference energy is much smaller, while the absolute discrepancy remains 9.39×10^{-14} . Both satisfy the 10^{-8} algebraic diagnostic tolerance. For 16 independent high-precision materials, all error matrices $C_{48}^{\text{eff}} - C_{\text{FOM}}^{\text{eff}}$ are positive definite; their minimum eigenvalue over the campaign is 3.17×10^{-4} , confirming the hierarchy in Eq. (17). The optimized 129-reference bound has zero strict Loewner violations and minimum eigenvalue 2.53×10^{-3} for the direct gap $U_{48} - (C_{48}^{\text{eff}} - C_{\text{FOM}}^{\text{eff}})$. Its scalar effectivity from Eq. (21) ranges from 2.86 to 25.51, with median 8.94; optimizing convex combinations of the 129 valid references selects one active reference and does not improve that median.

Equation (22) changes the task from bounding the full $r = 48$ error to bounding only the enriched $r = 168$ tail. That tail is a median 0.855% of the $r = 48$ error norm. Adding its valid two-kernel upper bound to the exact $C_{48}^{\text{eff}} - C_{168}^{\text{eff}}$ drop gives zero strict violations, a minimum Loewner gap 3.45×10^{-5} , and effectivity between 1.012 and 1.411, with median 1.098 and 95th percentile 1.388. Thus the median below five is met without SCM or an anisotropic reference. With one BLAS thread, median tail-estimator time over the 16 strict rows is 1.90 ms. A separate 4096-material benchmark

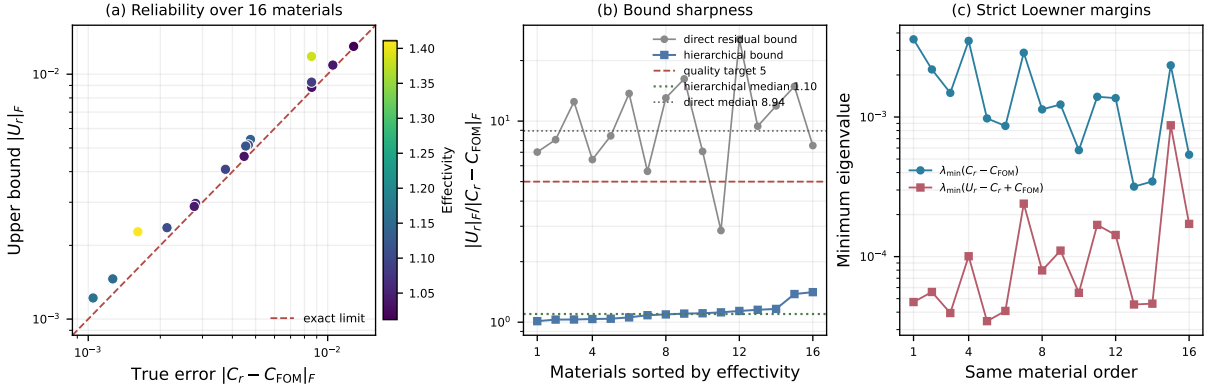


Figure 4: Campaign-level audit of the Schur bound. (a) Every hierarchical upper bound lies above the exact-limit diagonal. (b) The direct residual bound has median effectivity 8.94, whereas the exact reduced drop plus bounded enriched tail lowers it to 1.10, below the target value 5. (c) Positive minimum eigenvalues verify both $C_r^{\text{eff}} - C_{\text{FOM}}^{\text{eff}} \geq 0$ and $U_r^{\text{hier}} - (C_r^{\text{eff}} - C_{\text{FOM}}^{\text{eff}}) \geq 0$ for all 16 high-precision materials.

of the complete two-ROM bounded query takes 6.64 s in total, with median 1.53 ms and 95th percentile 1.95 ms. This sharp bridge controls the $r = 48$ output while using $r = 168$ as an auxiliary tail space. A separate, predeclared test of the deployed $r = 168$ operator added 20 materials selected only by the computable certificate width, producing an $r = 288$ auxiliary space. On 16 float64, rtol = 10^{-12} truths, its upper gap has zero strict violations. Over 4096 candidates, 3988 implied relative bounds satisfy 10^{-4} , with median, 95th-percentile, and maximum values 1.87×10^{-5} , 7.36×10^{-5} , and 4.09×10^{-4} . The auxiliary level does not sharpen the final-level effectivity, whose median is 13.38, and tightening the truth solve does not remove lower-hierarchy sign changes at the 10^{-5} error scale. We therefore retain $r = 168$, already below the independent validation target, as the deployed operator and do not use “Certified” in the title. Figure 4 shows the direct ablation and hierarchical result together. Figure 5 resolves the six Loewner eigenvalues through the $48 \rightarrow 168$ bridge for the most difficult material, while Table 5 summarizes the campaign.

7.3. Adaptive Sobol+POD convergence

Instead of prescribing a fixed material budget, we build the goal-oriented constitutive transfer operator by iteratively adding full-rank Sobol snapshots until the relative tensor error falls below a target tolerance. This adaptive procedure naturally sizes the reduced basis to the complexity of the geometry without requiring candidate surrogate evaluations.

Figure 6 shows the maximum relative Frobenius error evaluated on the 256 high-precision truth tensors as materials are added to the POD basis. The operator reaches the 2×10^{-5} target tolerance after 38 Sobol materials, producing a reduced basis of rank 228. Because the offline snapshots are solved sequentially on the GPU, the procedure does not saturate device memory. Crucially, the online dimension does not impact the constant online evaluation cost due to the exact algebraic factorization.

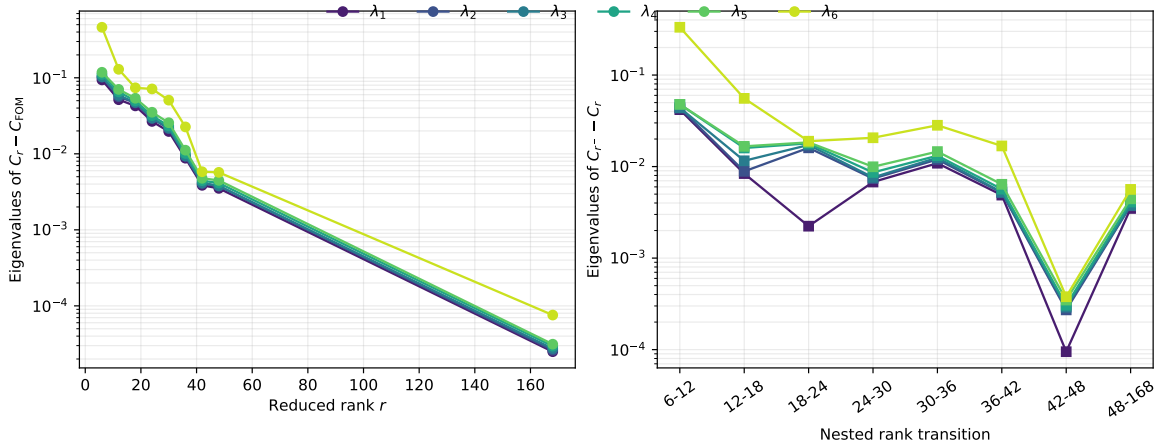


Figure 5: Six eigenvalues of $C_r^{\text{eff}} - C_{\text{FOM}}^{\text{eff}}$ and of differences between consecutive nested levels for the most difficult high-precision material. Positivity visualizes the Ritz/Schur Loewner hierarchy.

Table 5

High-precision direct and hierarchical bound study.

Quantity	Value
High-precision materials	16
Coarse/enriched ranks	48/168
Strict violations	0
Direct median effectivity	8.939
Hierarchical median effectivity	1.098
Hierarchical P95 effectivity	1.388
Median estimator [ms]	1.434
4096-query bounded sweep [s]	6.708
Minimum Loewner gap	3.449e-05
Production/auxiliary ranks	168/288
Final upper violations	0
Implied bounds $\leq 10^{-4}$	3988/4096
Final width median/P95	1.87e-05/7.36e-05
Final maximum width	4.09e-04
Final 4096-query sweep [s]	14.481

7.4. Sensitivity checks

The analytic sensitivities in Eq. (25) match internal finite differences of the ROM at near machine precision. Against full-order finite differences, the $r = 66$ center-material check has mean relative sensitivity error 4.35×10^{-4} and maximum 1.65×10^{-3} . The same $r = 66$ operator is less accurate for a derivative-problematic validation material, with mean 9.76×10^{-3} and maximum 2.91×10^{-2} , even though its tensor error is only 3.65×10^{-4} . Adding that material as one tangential block gives $r = 72$ and reduces the sensitivity errors to mean 2.27×10^{-4} and maximum 4.57×10^{-4} . Figure 7 shows this targeted reduction, and Table 6 reports the corresponding tensor and derivative errors.

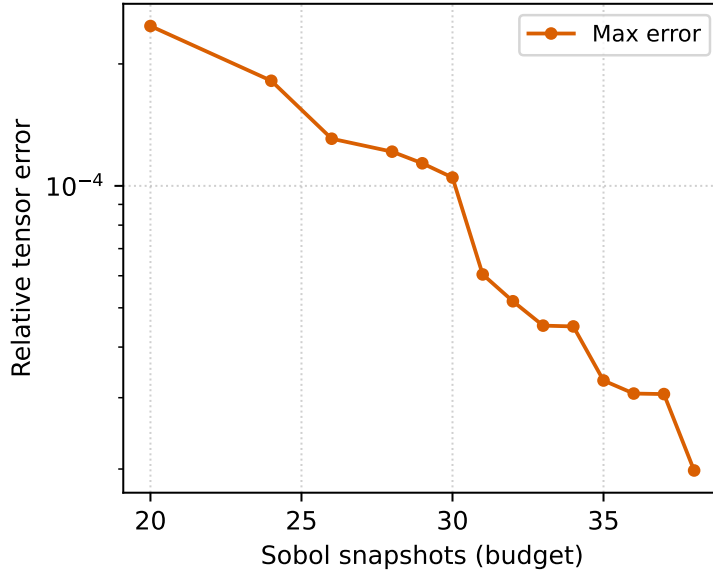


Figure 6: Adaptive full-rank Sobol+POD convergence on 256 untouched truth tensors. The horizontal dashed line marks the 2×10^{-5} target tolerance.

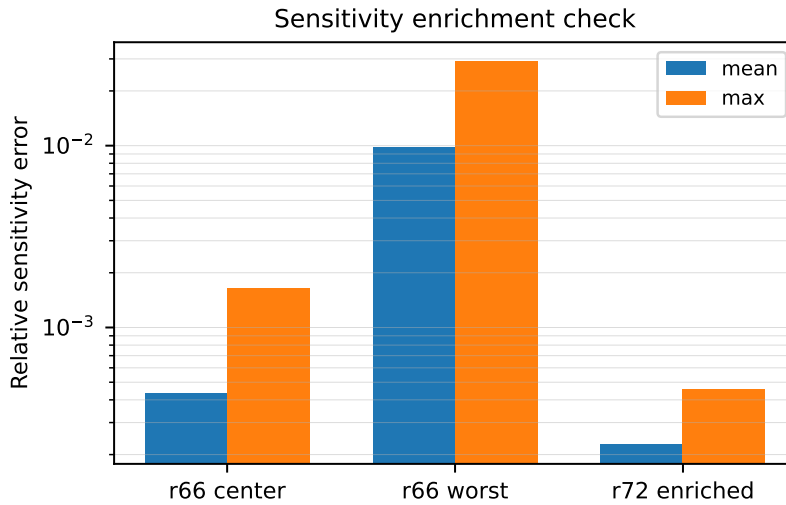


Figure 7: Full-order finite-difference sensitivity checks before and after targeted tangential enrichment.

7.5. Discretization and geometry scope

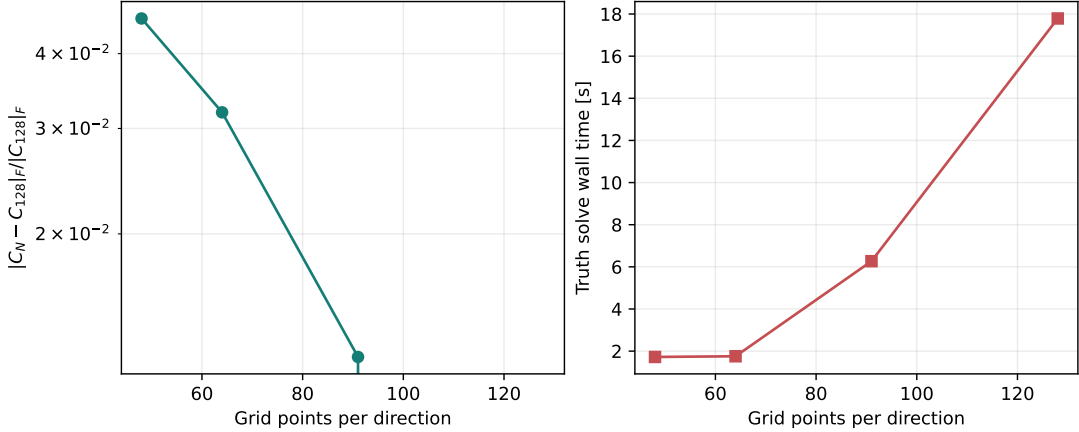
7.5.1. Voxel refinement on one continuous geometry

All 96 fibers of the baseline continuous geometry were rasterized without regeneration at $48^3, 64^3, 91^3, 128^3$. Relative to the finest tensor, the discretization errors are 4.58×10^{-2} , 3.19×10^{-2} , and 1.25×10^{-2} , respectively, as detailed in Figure 8 and Table 7. This separates voxel error from material-space ROM error: the 10^{-4} ROM threshold controls the transfer operator on a fixed discretization and does not imply a 10^{-4} continuum homogenization error.

Table 6

Reduced analytic sensitivity checks against full-order finite differences.

Case	Mean	Median	Max
r66 center	4.353e-04	1.550e-04	1.651e-03
r66 worst validation material	9.762e-03	8.477e-03	2.907e-02
r72 enriched same material	2.267e-04	2.227e-04	4.569e-04
r168 edge-auto check	9.407e-04	1.999e-04	5.406e-03

**Figure 8:** Discretization error and high-precision solve cost for the same continuous 96-fiber geometry rasterized at four grids.**Table 7**

Voxel-refinement truth solves on one continuous geometry.

N	N^3	V_f	rel. error vs. 128^3	solve [s]
48	22094	0.19978	4.579e-02	1.72
64	52196	0.19911	3.192e-02	1.76
91	150213	0.19933	1.246e-02	6.27
128	418156	0.19939	0.000e+00	17.79

An independent spherical-inclusion benchmark uses isotropic phases at 10% volume fraction and grids $32^3, 48^3, 64^3, 91^3$. The relative tensor error against 91^3 falls from 2.96×10^{-3} to 1.18×10^{-4} at 64^3 ; the refinement curve and bound margins are reported in Figure 9 and Table 8. Every tensor satisfies the full Mandel Voigt/Reuss Loewner bounds with positive margins, and the hydrostatic response lies inside the Hashin–Shtrikman bulk interval. The periodic sphere array has cubic, not isotropic, effective symmetry; consequently an isotropic scalar Hashin–Shtrikman shear interval is reported only as a diagnostic and is not used as an acceptance criterion.

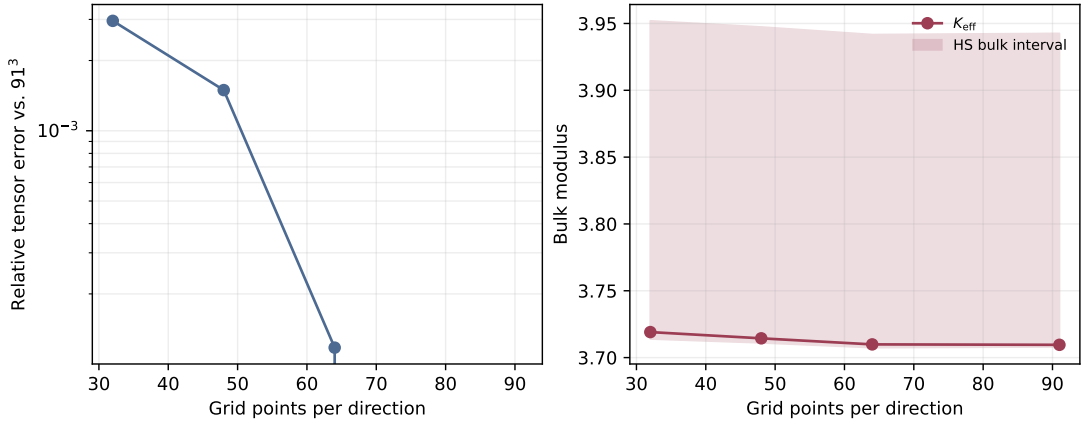


Figure 9: Spherical-inclusion refinement and Hashin–Shtrikman bulk interval.

Table 8

Inclusion refinement and matrix-valued Voigt/Reuss margins.

N	V_f	rel. error	min eig. above Reuss	min eig. below Voigt
32	0.10156	2.958e-03	1.766e-01	7.902e-01
48	0.10084	1.494e-03	1.731e-01	7.884e-01
64	0.09995	1.177e-04	1.696e-01	7.822e-01
91	0.10008	0.000e+00	1.686e-01	7.847e-01

Table 9

Acceptance and separation of the ten-geometry campaign.

Quantity	Value
Accepted geometries	10 / 10
Minimum design distance	1.038
Minimum realized-descriptor distance	0.758
Fixed grid	91^3

7.5.2. Ten distinct microstructures

A reproducible maximin design spans $V_f \in [0.10, 0.30]$, $AR \in [8, 25]$, diagonal orientation tensors, and clustering fraction in $[0, 0.70]$. Periodic Gaussian-mixture center proposals were added to SAM for the clustered cases. All ten RVEs satisfy the volume, orientation, and overlap gates. They also remain separated in the realized descriptor space formed from interface density, a Ripley statistic, discrepancy, fiber count, and A_2 ; the minimum normalized realized distance is 0.758. Figure 10 compares target and realized descriptors, Figure 11 displays the accepted RVEs, and Table 9 records their acceptance margins. This campaign validates geometry generation and prevents one unusually reducible RVE from being presented as universal evidence. Each accepted geometry defines a separate compiled operator; no transfer of one reduced basis between RVEs is assumed.

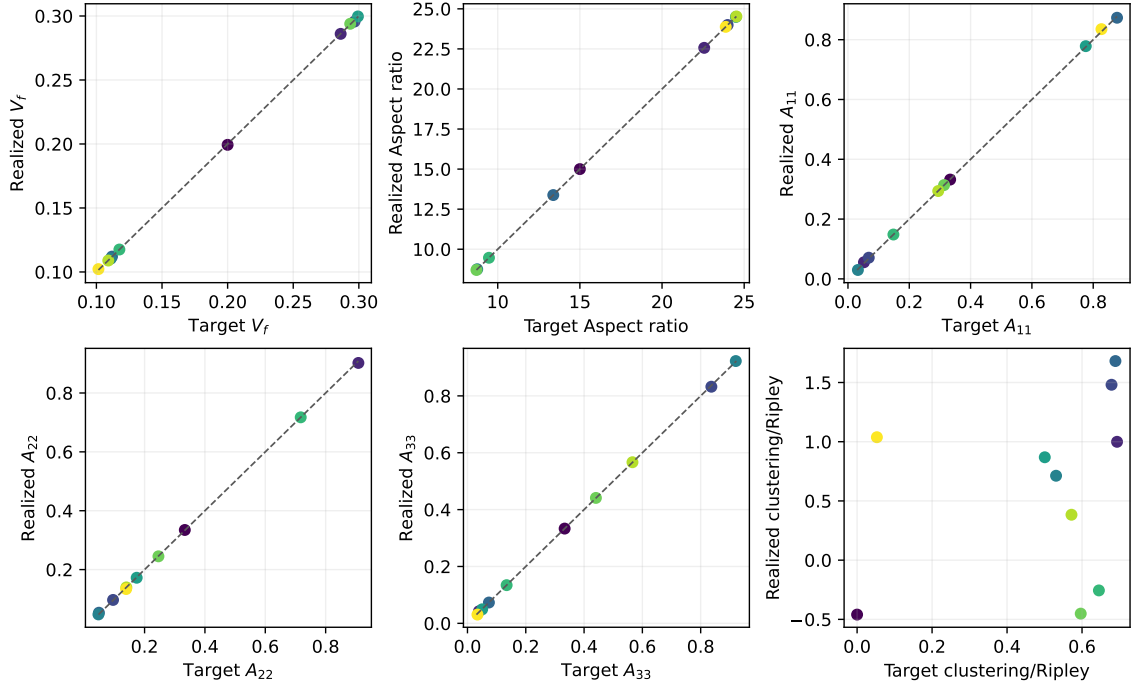


Figure 10: Target versus realized volume fraction, aspect ratio, orientation tensor components, and clustering response for ten accepted microstructures.

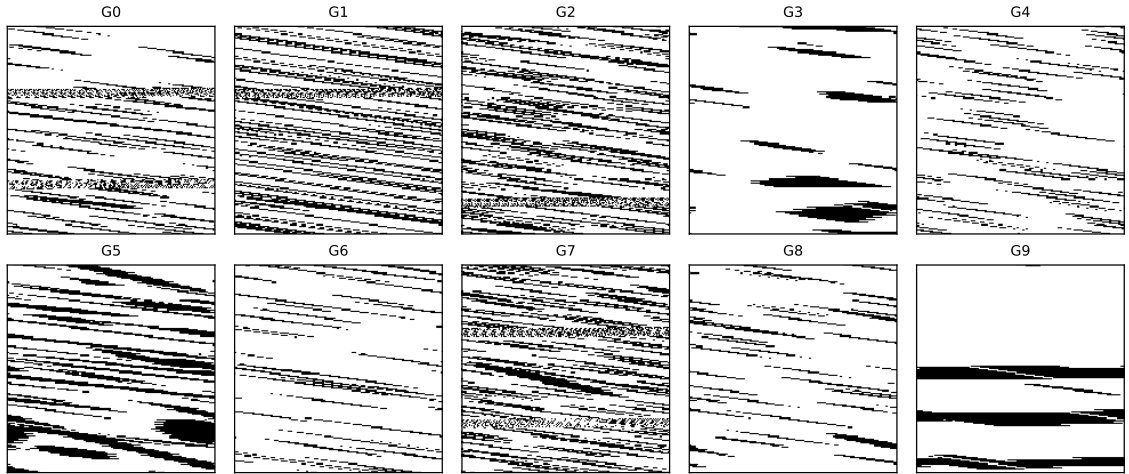


Figure 11: Central slices of the ten accepted 91^3 periodic RVEs.

7.6. Many-query applications

The preceding subsections establish approximation, error control, selection behavior, sensitivity accuracy, and the discretization scope. The screening, identification, and uncertainty-propagation applications test the other part of the research question: whether the compiled operator can replace new voxel solves once deployed inside a many-query workflow. They use the final fixed-geometry operator and add no full-order information online.

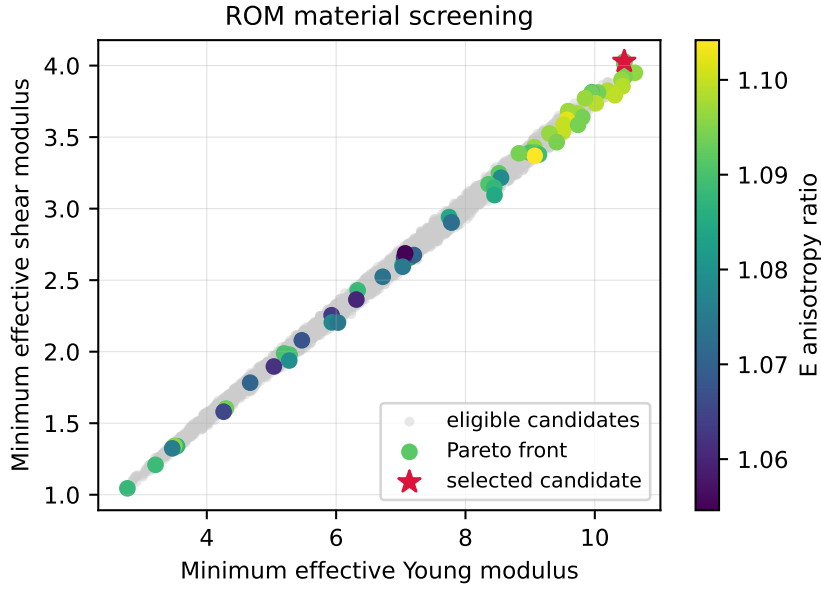


Figure 12: ROM-only material screening. Gray points are eligible candidates, colored points are the Pareto front, and the star is the selected best balanced-gain candidate.

Table 10

Best reduced-model candidate in the 4096-point material-screening study.

Quantity	Value
Candidate id	413
E_m, ν_m	4.2873, 0.3510
E_L, E_T, G_{LT}	383.581, 22.052, 25.861
ν_{LT}, ν_{TT}	0.2549, 0.3904
$E1, E2, E3$	10.4596, 10.4555, 11.4120
$G12, G13, G23$	4.4647, 4.0286, 4.0905
Balanced gain	1.552986
E anisotropy ratio	1.091481
Median ROM online time	1.024e-03
Full-order solves used	0

7.6.1. ROM-only material screening

The $r = 168$ operator was used for ROM-only screening over 4096 Sobol candidates. No full-order FFTHomPy solves were run during screening. All 4096 candidates were eligible under the prescribed constraints, and 64 were identified on the Pareto front. The selected candidate maximizes the balanced gain over the effective Young and shear moduli relative to the center material while keeping the reduced response stable. Median reduced query time is 0.138 ms with one BLAS thread. Figure 12 shows the screening cloud and Pareto set; Table 10 gives the selected material and its effective response.

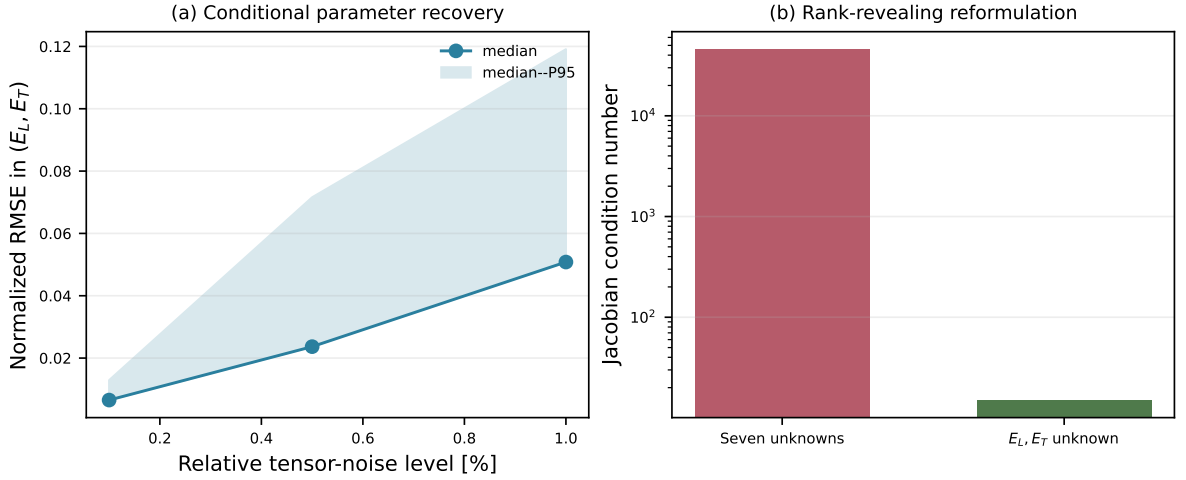


Figure 13: Sensitivity-guided inverse reformulation. (a) Median and 95th-percentile normalized recovery error when only E_L and E_T are unknown. (b) Restricting the inverse problem to those observable directions reduces the Jacobian condition number by about three orders of magnitude.

7.6.2. Inverse identification

The ROM sensitivities are first used as a rank-revealing diagnostic, following standard sensitivity-based practical-identifiability analysis [4], rather than assuming that all seven material constants can be inferred from one homogenized tensor. For validation material 94, the normalized 21×7 Jacobian is algebraically full rank but has condition number 4.58×10^4 ; its two weakest right-singular directions are dominated by the fiber Poisson ratios. Consequently, the seven-unknown problem is not used as the application claim. Its complete noise and regularization stress test is retained in Appendix D.

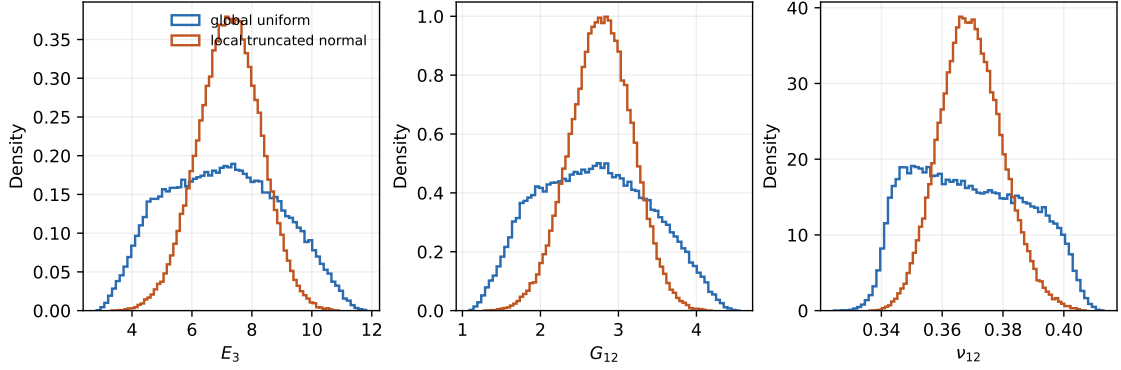
We instead pose a conditional identification problem in which matrix properties, G_{LT} , and the two fiber Poisson ratios are characterized independently, while E_L and E_T are inferred from the effective tensor. In normalized coordinates $z_I = (z_{E_L}, z_{E_T})$, the ROM-only problem is

$$\min_{z_I \in [0,1]^2} \frac{\|C_r^{\text{eff}}(z_I; z_{\text{known}}) - C^{\text{obs}}\|_F^2}{\|C^{\text{obs}}\|_F^2}. \quad (30)$$

The corresponding Jacobian condition number is approximately 15, versus 4.58×10^4 for seven unknowns. Across ten independent symmetric perturbations at each noise level, the median normalized RMSE in (E_L, E_T) is 0.0065, 0.0237, and 0.0508 for 0.1%, 0.5%, and 1% tensor noise, respectively; the corresponding 95th percentiles are 0.0129, 0.0716, and 0.119. All 30 four-start optimizations converge and require no new full-order solve. Figure 13 and Table 11 report the recovery and conditioning. The result supports conditional identification of observable parameters, not unrestricted recovery of seven individual constants.

Table 11Conditional (E_L, E_T) identification under tensor noise.

Noise [%]	Parameter RMSE	P95	Tensor error
0.1	0.007	0.013	2.82e-04
0.5	0.024	0.072	1.40e-03
1.0	0.051	0.119	2.09e-03

**Figure 14:** Global-uniform and local truncated-normal uncertainty propagation for representative Young, shear, and Poisson outputs.**Table 12**

Selected statistics from 100 000 samples per input law.

Distribution	output	mean	std.	P05	P95
global	E3	7.0355	1.8432	4.1517	10.1020
global	G12	2.6946	0.7009	1.5999	3.8676
global	nu12	0.3692	0.0181	0.3427	0.3994
local	E3	7.2482	1.0447	5.5046	8.9493
local	G12	2.7793	0.3971	2.1165	3.4259
local	nu12	0.3696	0.0103	0.3531	0.3872

7.6.3. Uncertainty propagation

The final operator was queried for 100 000 independent global-uniform materials and 100 000 independent local truncated-normal materials. The local law is centered at the midpoint of each of the seven physical intervals with $\sigma = (b - a)/6$, truncated to the admissible box. No predicted effective tensor lost positive definiteness. The 200 000 reduced solves take 46.2 s with one BLAS thread. The local law preserves similar central values but reduces the standard deviations; for example, the standard deviation of E_3 falls from 1.843 to 1.045 and that of G_{12} from 0.701 to 0.397. At 50 000 samples, the largest mean change relative to the final 100 000-sample statistics is 7.53×10^{-4} ; the largest 5th/95th-percentile changes are 5.54×10^{-4} and 2.71×10^{-3} , respectively. The saved convergence table therefore checks both moments and distribution tails rather than treating 100 000 as an unsupported sample count. Figure 14 compares the output distributions and Table 12 gives their selected moments.

7.7. Claim boundaries

The guarantee concerns ROM reduction relative to the discrete FFT problem. It does not absorb the separately measured voxel error or transfer one basis between different microstructures; each geometry is compiled independently. Likewise, the seven physical material inputs are retained although global elastic-scale homogeneity admits the six-dimensional quotient described in Appendix E.

The causal comparator supports no universal selector ranking, while the tuned black-box control isolates the value of the intrusive Schur construction in this data regime. A cross-code FEM comparison is outside the present scope; solver credibility is instead established by the homogeneous, laminate, Mandel, equilibrium, compatibility, Hill–Mandel, voxel-refinement, and inclusion-refinement checks, including classical Voigt/Reuss and hydrostatic Hashin–Shtrikman bounds.

8. Conclusions

We formulated a goal-oriented constitutive transfer framework for parametric computational homogenization. It combines six-load physical tangential enrichment, constitutive Hermite interpolation, a Ritz/Schur error identity, and a transverse/longitudinal two-kernel estimator whose online complexity is independent of voxel count at fixed rank. The FFTHomPy/CuPy study closes solver convergence, basic energetic identities, dense/direct estimator agreement, and a zero-violation discrete bound validation. The screening, identification, and 200 000-query UQ studies in Section 7.6 demonstrate the intended many-query use. Inversion is restricted by a rank-revealing sensitivity analysis to the conditionally observable pair (E_L, E_T) ; the ill-conditioned seven-unknown case is retained as a negative control. The bound is reliable and sharp after hierarchical tail correction; its guarantee remains relative to the discrete FFT truth problem. The causal campaign and the time-matched Sobol controls reject universal material-selection dominance: fixed Sobol+POD can give lower raw tensor error when it receives the same offline time. Selection is therefore an ablation, whereas the transferable contribution is the structure-aware operator, its Schur diagnostics, and its goal-oriented outputs. The black-box control isolates the benefit of that intrusive construction. Voxel error remains separate from ROM error, and every new geometry requires its own offline compilation.

A. Affine constitutive coefficients

For the isotropic matrix,

$$\mu_m = \frac{E_m}{2(1 + \nu_m)}, \quad \lambda_m = \frac{E_m \nu_m}{(1 + \nu_m)(1 - 2\nu_m)}.$$

The local transversely isotropic fiber stiffness is expressed in Mandel coordinates using five independent coefficients $(c_{TT}, c_{TT}^x, c_{LT}, c_{LL}, g_{LT})$. For a voxel whose local fiber direction is $a(x)$, the local tensor basis is rotated into the global frame before being accumulated into $K_q(\mathcal{G})$, $B_q(\mathcal{G})$, and $D_q(\mathcal{G})$. The implementation uses Mandel shear normalization consistently in the full-order tensors, reduced tensors, residual indicators, and generalized eigenvalue calculations for β .

B. Local coercivity factor and reference selection

Residual bounds obtained by comparing the energy operator with a coercive reference are standard in certified RB methodology [22, 23]. Here the argument is written in Loewner form because the output is the full effective tensor.

If M is symmetric positive definite and

$$K \geq \beta M, \quad \beta > 0,$$

then

$$K^{-1} \leq \beta^{-1} M^{-1}.$$

Combining this with the classical identity (16) gives

$$0 \leq C_r^{\text{eff}} - C_{\text{FOM}}^{\text{eff}} \leq \beta^{-1} R^T M^{-1} R.$$

For the voxelwise elastic operator, it is sufficient to impose the local generalized eigenvalue condition

$$C(x; \xi) \geq \beta(\xi) C_0(\xi) \quad \text{for every voxel } x.$$

With an isotropic reference C_0 ,

$$\beta(\xi) = \min\{\lambda_{\min}(C_m, C_0), \lambda_{\min}(C_f, C_0)\}.$$

The implementation evaluates these generalized spectra for 129 reference shapes, multiplies β by $1 - 10^{-10}$, and adds a conservative iterative-truth guard based on the recorded full-order relative residual. This guard accounts for numerical comparison uncertainty; it is not a certificate of continuum discretization error.

C. Two-kernel offline–online factorization

Let $\hat{\tau}(k)$ be the Fourier transform of an affine residual-stress atom and $n = k/\|k\|$. Its traction is $t = \hat{\tau}n$, with $t_L = (t \cdot n)n$ and $t_T = t - t_L$. For an isotropic reference medium, Parseval's identity yields

$$\langle r, M^{-1}r \rangle = \sum_{k \neq 0} \left(\frac{\|t_T(k)\|^2}{\mu_0} + \frac{\|t_L(k)\|^2}{\lambda_0 + 2\mu_0} \right).$$

The isotropic Fourier Green operator and its transverse/longitudinal split are classical ingredients of FFT homogenization [20, 26].

Proof of Proposition 1. For a fixed reduced coordinate matrix $Y = Y_r(\xi)$, define the six-column affine residual atom

$$S_p(Y) = B_p - K_p V_r Y.$$

Then $R = \sum_p \gamma_p S_p(Y)$, and bilinearity gives, for $\alpha \in \{T, L\}$,

$$R^T P_\alpha R = \sum_{p,q} \gamma_p \gamma_q S_p(Y)^T P_\alpha S_q(Y). \quad (31)$$

Because $P_\alpha = P_\alpha^T$, expanding one summand yields

$$\begin{aligned} S_p(Y)^T P_\alpha S_q(Y) &= B_p^T P_\alpha B_q - B_p^T P_\alpha K_q V_r Y \\ &\quad - Y^T V_r^T K_p^T P_\alpha B_q + Y^T V_r^T K_p^T P_\alpha K_q V_r Y \\ &= BB_{pq}^\alpha - BK_{pq}^\alpha Y - Y^T (BK_{qp}^\alpha)^T + Y^T K K_{pq}^\alpha Y. \end{aligned} \quad (32)$$

Substitution of Eq. (32) into Eq. (31) gives Eq. (14); substitution of the classical orthogonal split (12) gives Eq. (15).

Thus the dense and direct-Fourier forms are algebraically identical.

Each BB , BK , and KK block contains only $n_g \times n_g$, $n_g \times r$, and $r \times r$ arrays, respectively. Once those blocks have been compiled, the pairwise contractions require $O(Q^2 r^2 + Q^2 r n_g)$ arithmetic for fixed $n_g = 6$ and $O(Q^2 r^2)$ storage. Every FFT and voxel sum occurs in the construction of the blocks, so neither online quantity contains N_{vox} . $\square$

D. Seven-parameter identifiability stress test

For completeness, all seven normalized material coordinates were first treated as unknowns in

$$\min_{z \in [0,1]^7} \frac{\|C_r^{\text{eff}}(z) - C^{\text{obs}}\|_F^2}{\|C^{\text{obs}}\|_F^2} + \lambda_{\text{reg}} \|z - \tfrac{1}{2} \mathbf{1}\|_2^2, \quad \lambda_{\text{reg}} \in \{0, 10^{-6}, 10^{-4}, 10^{-2}\}.$$

Ten independent symmetric perturbations were used at each of the 0.1%, 0.5%, and 1% tensor-noise levels. All 120 optimizations converged, but the best median normalized seven-parameter RMSE values were 0.183, 0.227, and 0.239, respectively. At 0.1% noise, for example, the recovered-tensor error remained only 4.44×10^{-4} . Small output residuals therefore do not imply stable recovery along weak Jacobian directions. This negative control motivates the conditional problem in Eq. (30); it is not counted as a successful seven-parameter identification result. The complete 120-run table is retained in the supplementary campaign archive.

E. Elastic-scale homogeneity

If every constituent stiffness is multiplied by $s > 0$, then K , B , D , $C_{\text{FOM}}^{\text{eff}}$, and C^{eff} scale by s , while the fluctuation response $K^{-1}B$ is invariant. Hence one physical modulus could be factored out and the shape of the transfer map represented by six nondimensional ratios. This study deliberately retains the seven physical inputs because its sampling, validation, inverse problem, and user-facing material interface were defined in those variables. No dimensionality-reduction novelty is claimed.

F. Reproducibility

The CMAME figures and tables are generated from campaign manifests with

```
pythonscripts/cmame_generate_paper_assets.py.
```

The source campaigns are under `results/cmame_method`; generated files and hashes are recorded in `paper/cmame_assets_manifest.json`. The manuscript is compiled from `paper` using:

```
pdflatex-interaction=nonstopmodemain.tex
bibtexmain
pdflatex-interaction=nonstopmodemain.tex
pdflatex-interaction=nonstopmodemain.tex
```

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The source code, campaign manifests, and derived data needed to reproduce every reported table and figure are provided in the accompanying supplementary archive. Baseline artifacts are separated from the new methodological campaigns, and each generated manuscript asset is recorded with its source path and SHA-256 hash.

CRediT authorship contribution statement

Kevin Vidal: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Victor Alulema:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – review and editing. **Victor Berrazueta:** Validation, Resources, Writing – review and editing. **Patrick Vandewalle:** Methodology, Supervision, Resources, Writing – review and editing. **Esteban Valencia:** Supervision, Project administration, Resources, Writing – review and editing.

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